Example 1: Single Linkage with City Block Distance

Question. For the one-dimensional data set **{7, 10, 20, 28, 35}**, perform agglomerative clustering with single linkage and plot the dendrogram to visualize it using City block distance.

CITY BLOCK (MANHATTAN) DISTANCE

$d(x, y) = \Sigma\ |x_i - y_i|$. In one dimension this is simply **$d(x, y) = |x - y|$**.

SINGLE LINKAGE

The distance between two clusters is the **smallest** distance between any member of one and any member of the other.

Procedure: start with each value as its own cluster, merge the two closest clusters, update distances with the single-linkage rule, and repeat until one cluster remains.

Solution

STEP 0 · DISTANCE MATRIX

	7	10	20	28	35
7	0				
10	3	0			
20	13	10	0		
28	21	18	8	0	
35	28	25	15	7	0

```
d(7,10)     |7 − 10| = 3
d(7,20)     |7 − 20| = 13
d(7,28)     |7 − 28| = 21
d(7,35)     |7 − 35| = 28
d(10,20)    |10 − 20| = 10
d(10,28)    |10 − 28| = 18
d(10,35)    |10 − 35| = 25
d(20,28)    |20 − 28| = 8
d(20,35)    |20 − 35| = 15
d(28,35)    |28 − 35| = 7
```

STEP 1

	7	10	20	28	35
7	0				
10	**3**	0			
20	13	10	0		
28	21	18	8	0	
35	28	25	15	7	0

Smallest entry = **3** between 7 and 10. Merge them into **{7,10}** at height 3.

Update distances (single linkage = minimum):

```
d({7,10}, 20)   min(d(7,20), d(10,20)) = min(13, 10) = 10
d({7,10}, 28)   min(d(7,28), d(10,28)) = min(21, 18) = 18
d({7,10}, 35)   min(d(7,35), d(10,35)) = min(28, 25) = 25
```

STEP 2

	{7,10}	20	28	35
{7,10}	0			
20	10	0		
28	18	8	0	
35	25	15	**7**	0

Smallest entry = **7** between 28 and 35. Merge them into **{28,35}** at height 7.

Update distances (single linkage = minimum):

```
d({28,35}, {7,10}) min(d(28,7), d(28,10), d(35,7), d(35,10)) =
min(21, 18, 28, 25) = 18
d({28,35}, 20)  min(d(28,20), d(35,20)) = min(8, 15) = 8
```

STEP 3

	{7,10}	20	{28,35}
{7,10}	0		
20	10	0	
{28,35}	18	**8**	0

Smallest entry = **8** between 20 and {28,35}. Merge them into **{20,28,35}** at height 8.

Update distances (single linkage = minimum):

`d({20,28,35}, {7,10}) min(d(20,7), d(20,10), d(28,7), d(28,10), d(35,7), d(35,10)) = min(13, 10, 21, 18, 28, 25) = 10`

STEP 4

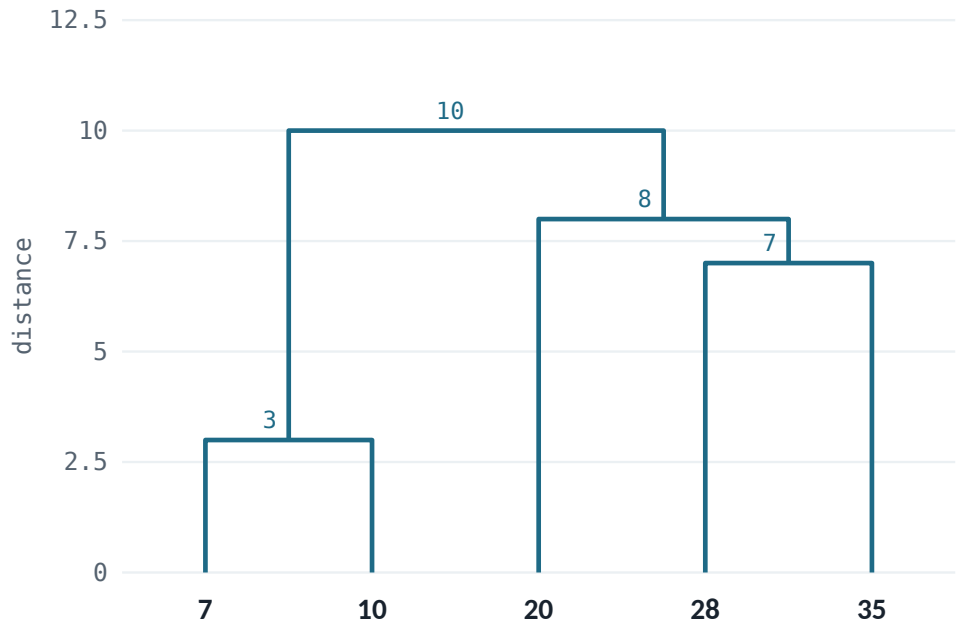
	{7,10}	{20,28,35}
{7,10}	0	
{20,28,35}	**10**	0

Smallest entry = **10** between {7,10} and {20,28,35}. Merge them into **{7,10,20,28,35}** at height 10.

Only one cluster remains. Clustering is complete.

Summary of merges

MERGE	CLUSTERS JOINED	HEIGHT
1	7 + 10	3
2	28 + 35	7
3	20 + {28,35}	8
4	{7,10} + {20,28,35}	10



Dendrogram · single linkage, City block distance

Interpretation

- A cut between heights 8 and 10 gives **2 clusters**: {7, 10} and {20, 28, 35}.
- A cut between heights 7 and 8 gives **3 clusters**: {7, 10}, {20} and {28, 35}.
- Single linkage uses the nearest points between clusters. That is why 20 joins {28, 35} at height 8: its distance to 28 is 8, while its nearest point in the other cluster, 10, is 10 away.